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Adaptive Regulation of Micro Closed-Loop Space Station Ecosystems Using Proximal Policy Optimization

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Abstract

Long-term orbital missions require highly independent life-support systems, maintaining ecological stability under limited external support and subsystem attenuation. Conventional strategies are typically based on fixed threshold rules or deterministic mass-balance models and cannot adapt to disturbances or multi-objective trade-offs between resource safety and environmental load. This paper proposes a reinforcement learning framework for managing micro closed-loop ecological systems under reliability decay and environmental constraints. A reinforcement learning controller based on Proximal Policy Optimization (PPO) is trained to learn stable regulation policies with random disturbances. A dynamic state model is constructed to describe the coupled and time-sequential elements, ranging from water storage, oxygen concentration, energy reserve, subsystem reliability to ecological load. Degradation propagation is integrated into the transition dynamics through reliability-dependent recycling efficiency. Simulation experiments demonstrate that the proposed policy maintains all resource variables within the safety margins while reducing ecological imbalance compared with traditional strategies. Consequently, the proposed framework provides an expandable control architecture for independent regulation of space station life-support systems.

Keywords: Proximal Policy Optimization (PPO); Micro closed-loop space station ecosystems; Sensitivity analysis

Introduction

Long-term space station missions require reliable ecological management systems to maintain resource cycles under strict constraints, ranging from mass, energy to reliability [2],[4],[5],[6]. Different from short-term missions, long-term space station operations cannot rely on frequent external support, but only depend on internal micro closed-loop life-support systems to secure the environmental stability. Disturbance in one subsystem may propagate through resource flows, which may cause a series of impacts across the overall ecological system. Therefore, maintaining stable control of integrated cycles is a critical challenge for long-term missions [9].

Traditional life-support controlling strategies are typically based on fixed threshold control [3],[5]. These methods assume predictable subsystem performance. They only respond when system variables approach predefined safety limits. However, subsystem attenuation, random disturbances, and environmental fluctuations become increasingly significant during long-term operations [5]. Therefore, deterministic control strategy model cannot effectively react to nonlinear or urgent issues between dynamics and reliability. Consequently, rule-based and fixed strategies may lead to inefficient resource allocation or delayed responses, causing ecological imbalance.

Recent advances in reinforcement learning have provided effective tools for independent decision-making in complex systems. In this study, a reinforcement learning framework for micro closed-loop space station ecosystems is developed using Proximal Policy Optimization (PPO). A compact state-space model is constructed to represent the integrated elements of water storage, oxygen concentration, energy reserve, subsystem reliability, and ecological load. The proposed model enables adaptive regulation policies, maintaining ecological stability under disturbances and subsystem degradation. In addition to regulation performance, this study further examines policy response characteristics and discusses deployment-oriented computational considerations for autonomous onboard control.

Micro Closed-Loop Ecosystem Modeling

State Representation

The micro closed-loop life-support system is represented by a compact state vector, which captures the most essential resource, reliability, and environmental elements. At decision step t, the system state is defined as

$$x_t = [W_t, O_t, E_t, R_t, L_t]^T \quad (1)$$

W_t denotes the normalized water storage level, O_t represents the normalized oxygen concentration, E_t is regarded as the available energy reserve, R_t denotes the overall reliability of the integrated subsystems, and L_t is the ecological load index associated with internal environmental stress.

Water Resource Dynamics

Water balance is determined by crew consumption, recovery, and external supply [5],[8]. P represents the crew population size, ω is the average water consumption rate, $\eta_w(R_t)$ is the recycling efficiency with the current subsystem reliability level, and $u_{w,t}$ denotes the supplemental water supply applied at step t. The corresponding sequential state can be expressed as

$$W_{t+1} = W_t - \omega P + \eta_w(R_t)\,\omega P + u_{w,t} \quad (2)$$

The second term represents net water demand from crew activity. And the third term reflects the proportion of consumed water recovered from the recycling subsystem. The last term is introduced as a controllable input, which compensates for temporary imbalance when recycling performance deteriorates.

Oxygen Balance

Oxygen evolution is modeled in a similar way as the water, considering metabolic consumption and regeneration from life-support system. In oxygen update is written as

$$O_{t+1} = O_t - \rho P + \eta_o(R_t)\,\rho P \quad (3)$$

where ρ is the average oxygen consumption rate, which is based on the previous statistic and $\eta_o(R_t)$ represents the oxygen regeneration efficiency under the current reliability state. This equation captures the fact that oxygen recovery capacity is also affected by subsystem attenuation. When reliability decreases, the effective regeneration rate declines, increasing the risk of ecological imbalance.

Energy Balance

Energy is treated as a shared supporting resource required for recycling and regulation processes. c_w and c_r represents the energy coefficients related to water supply and recycling regulation, respectively. $u_{r,t}$ is considered as the recycling control intensity, and g_t represents the available energy input at step t. The energy reserve evolves according to

$$E_{t+1} = E_t - c_w u_{w,t} - c_r u_{r,t} + g_t \quad (4)$$

Stabilizing the system requires energy expenditure, and intense control actions may reduce operational flexibility in the future when the energy storage is insufficient.

Reliability Modeling

Long-term operation undoubtedly leads to subsystem attenuation [5],[8]. To capture it, the overall reliability of the ecological system can be modeled as

$$R_t = \sum_{i=1}^{N} \omega_i\, r_{i,t}, \sum_{i=1}^{N} \omega_i = 1 \quad (5)$$

where $(r_{i,t})$ is the reliability of subsystem (i) at decision step (t), (ω_i) is its weighting coefficient, and (R_t) represents the overall reliability of the life-support subsystems.

$$R_{t+1} = R_t - \alpha R_t + \epsilon_t \quad (6)$$

In which α denotes the baseline attenuation rate and ϵ_t represents random disturbances, including minor recovery effects, temporary fluctuations, or rare shocks. This equation provides a simplified approximation of cumulative subsystem degradation.

To reflect the influence of reliability on recycling performance, the water recovery efficiency is assumed to satisfy

$$\eta_w(R_t) = \eta_{w0} R_t \quad (7)$$

where η_{w0} is the nominal recycling efficiency in ideal conditions. This relation means that degradation in subsystem reliability directly reduces recovery effectiveness, thereby integrating long-term equipment condition with short-term resource stability.

Ecological Load Evolution

In addition to direct resource balance, the internal environmental burden of the habitat must also be considered. Ecological load is

used here as an accumulated indicator of environmental stress caused by persistent resource and operational inefficiency. Its evolution can be modeled as

$$L_{t+1} = L_t + \beta_w(W_t - W^*)^2 + \beta_o(O_t - O^*)^2 + \beta_e(E_t - E^*)^2 \quad (8)$$

W^*, O^*, and E^* represents the nominal target values of water, oxygen, and energy, respectively, and β_w , β_o , and β_e are nonnegative weighting coefficients. This expression represents the persistent difference from ideal operating conditions and allows long-term imbalance to accumulate gradually over time.

Reinforcement Learning Formulation for Ecological Regulation

The life-support regulation problem is constructed as a reinforcement learning task for sequential decision-making under uncertainty[2]. At each decision step, the monitor observes the current state and determines how much regulation effort should be allocated to water support, energy redistribution, and recycling adjustment.

State Space

Specifically, the state at step t is defined as

$$s_t = x_t = [W_t, O_t, E_t, R_t, L_t]^T \quad (9)$$

Action Space

The regulation action is represented by a continuous control vector

$$a_t = [u_{w,t}, u_{e,t}, u_{r,t}]^T \quad (10)$$

Transition Dynamics

Given the current state s_t and control action a_t, the ecological system evolves according to the coupled nonlinear dynamics conditions defined previously. The corresponding transition equation can be written in compact form as

$$s_{t+1} = f(s_t, a_t) + \xi_t \quad (11)$$

In which $f(\cdot)$ represents the deterministic component of the ecological dynamics and ξ_t denotes random rare disturbances.

Safety Constraints

To ensure safe operation of the micro-closed loop ecosystem, the regulation process is subject to hard ecological constraints. These constraints are imposed on the primary resource variables as

$$W_t \geq W_{\min}, \quad O_{\min} \leq O_t \leq O_{\max}, \quad E_t \geq E_{\min} \quad (12)$$

$W_{\min}$, $O_{\min}$, $O_{\max}$, and $E_{\min}$ represents the minimum safety bounds. Violation of these conditions indicates that the ecological system has entered an unsafe and unstable operating region.

Reward Function

The reward function is designed to represent both ecological imbalance and unsafe control behavior. It is regarded as

$$r_t = -\lambda_1 L_t - \lambda_2 \Phi(s_t, a_t) \quad (13)$$

The coefficient λ_1 controls the relative importance of long-term ecological stability, while λ_2 determines the severity of penalties imposed on unsafe or inefficient control behavior.

Policy Optimization Using Proximal Policy Optimization

To solve the sequential ecological regulation problem defined in the previous section, a reinforcement learning method based on PPO is adopted. In contrast, PPO is well suited for this setting because it supports continuous action spaces and provides stable policy updates through a clipped objective.

Let $(\pi_\theta(a_t|s_t))$ denote the PPO policy and let $(\widehat{A_t})$ denote the estimated advantage function at decision step (t)[1]. The PPO objective is written as

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_{\mathbb{t}}\left[\min\left(r_t(\theta)\widehat{A_t},\ \text{clip}(r_t(\theta),\ 1-\epsilon,\ 1+\epsilon)\widehat{A_t}\right)\right] \quad (14)$$

In which $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio between the updated policy and the previous policy, and ϵ is the clipping coefficient that limits the magnitude of policy change during each update. By constraining the update step in this way, PPO reduces the risk of destructive parameter shifts and improves training stability under uncertain transitions.

The reinforcement learning controller consists of a policy network and a value network. Rectified Linear Unit activation is used in the hidden layers, and the policy is optimized using the Adam optimizer. Generalized Advantage Estimation is employed to reduce variance in the gradient update [1]. During training, the initial ecological states are randomly sampled within admissible operating ranges, and each episode evolves according to the stochastic transition model defined in previous section . At every decision step, the policy outputs sequential actions for accumulated water support, energy allocation, and recycling intensity. The resulting trajectories are then used to evaluate returns and update the policy iteratively until convergence. The training process converges steadily, as illustrated in Figure 1.

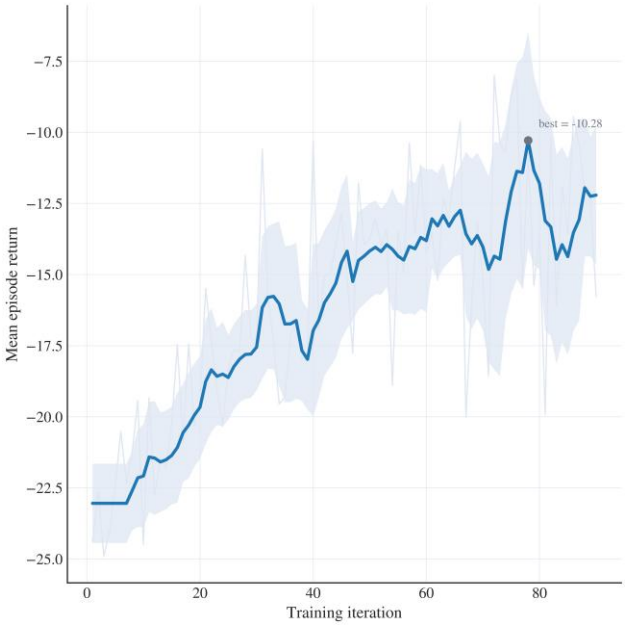


Figure 1. Training reward convergence during PPO optimization.

A practical configuration is to use a learning rate of 3×10^{-4}, a discount factor of 0.99, a clipping coefficient of 0.2, and a mini-batch size of 64. The maximum number of training iterations may be set to 1.0×10^{5} or higher depending on convergence behavior.

Simulation and Results

Experimental Setup

Simulation experiments were conducted over a horizon of 500 decision steps in a bid to evaluate the long-term regulation performance of the proposed method under random degradation. At the beginning of each episode, the state variables were initialized within admissible operating ranges around the nominal equilibrium point, so that the system started from a feasible but not perfectly balanced ecological condition. During simulation, subsystem reliability decayed progressively according to the degradation model introduced in previous section, while random disturbances were injected into the transition dynamics to represent uncertainty in recovery efficiency, energy availability, and environmental fluctuation.

The PPO policy was trained using the parameter settings reported in the previous section. The baseline controller was implemented using fixed safety thresholds and rule-based corrective actions, which provides a natural reference because such logic is widely used in practical ecological regulation. All simulations and PPO training were implemented in Python 3.10 with PyTorch 2.2 and CUDA 12.1 on an NVIDIA RTX 4090 GPU. Each experiment was repeated over 50 independent runs with different random seeds, and the reported results are averages over these runs. The present study is a simulation-based proof of concept rather than a flight-qualified validation campaign. No operational telemetry from closed-loop life-support platforms was available in the current stage, and no hardware-in-the-loop testbed was incorporated. Therefore, the reported results verify the control feasibility of the proposed PPO framework under structured stochastic disturbances, but do not yet establish full engineering readiness. A practical validation pathway would include disturbance replay using telemetry-like subsystem traces, followed by hardware-in-the-loop evaluation of policy latency, memory footprint, and closed-loop stability on resource-constrained onboard processors.

Resource Stability

The PPO-based policy maintained all major resource variables within admissible operating ranges throughout the simulation horizon. Compared with the threshold-based controller, the PPO-based policy exhibited smaller fluctuation and smoother recovery behavior after disturbances. As shown in Figure 2, the PPO-based controller maintains smoother water, oxygen, and energy trajectories than the threshold-based baseline. This difference is especially evident in the water and oxygen trajectories, where the baseline controller tended to react only after the state approached a safety boundary, whereas the learned policy adjusted regulation intensity earlier and avoided large excursions from the nominal operating point. The proposed policy reduced water storage variation by approximately 42%, oxygen concentration fluctuations were moderately reduced, and energy variance by 25% relative to the threshold-based strategy.

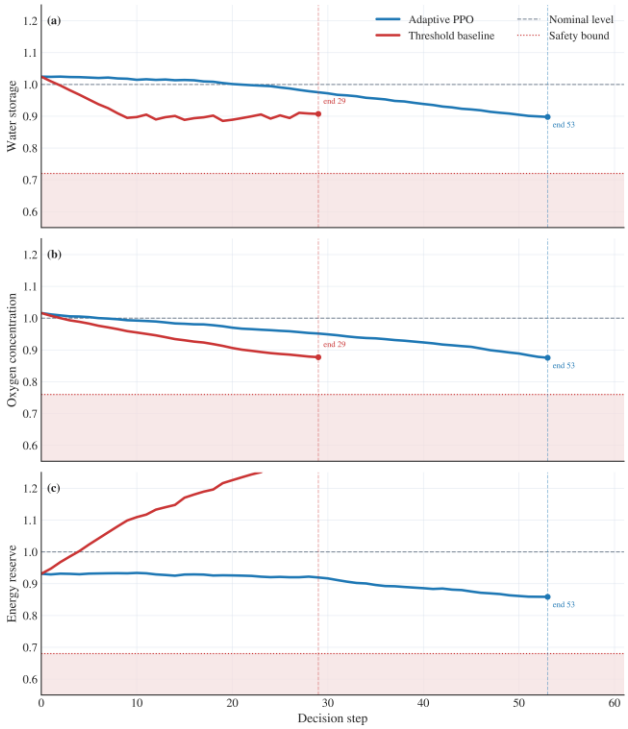


Figure 2. Resource state trajectories under adaptive policy and threshold control.

Ecological Load Suppression

Figure 3 shows that the ecological load grows more slowly under the PPO-based policy than under the threshold-based controller.

Under the PPO-based reinforcement learning policy, ecological load accumulation was reduced by approximately 38% compared with the threshold-based controller. This improvement indicates that the PPO policy was able to coordinate resource recovery and energy expenditure more effectively over time, instead of relying on short-term corrective actions. In particular, the learned controller tended to suppress deviation earlier, which limited the cumulative growth of environmental burden across the episode.

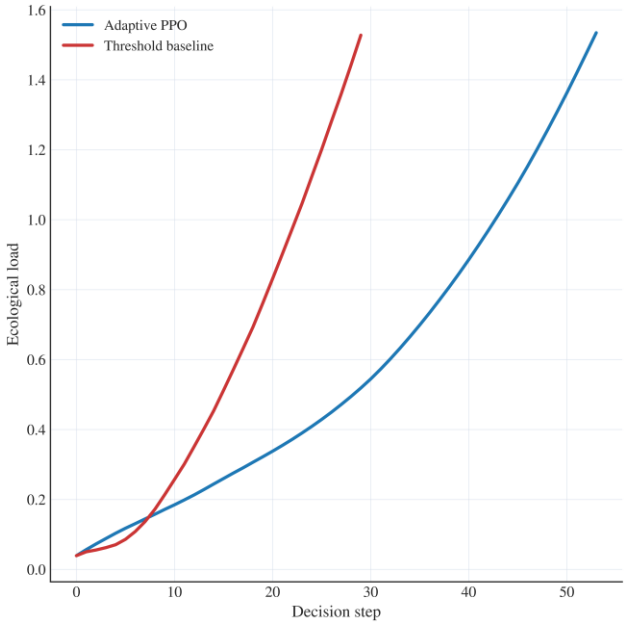


Figure 3. Ecological load evolution under adaptive PPO regulation and threshold-based control.

Robustness Under Degradation

To evaluate robustness, the degradation coefficient was varied across multiple scenarios while keeping the control structure unchanged. This experiment was designed to test whether the learned policy remained effective when the system worsens faster than the ideal case. The results show that the PPO-based adaptive policy preserved stable operation over a wider degradation range than the threshold-based baseline. This trend is evident in Figure 4, which compares system performance across multiple degradation scenarios. Even when degradation was accelerated, the learned policy continued to allocate regulation effort in a balanced manner and prevented abrupt ecological collapse.

An important result is that the proposed strategy extended safe operational duration by nearly 90% under accelerated degradation.

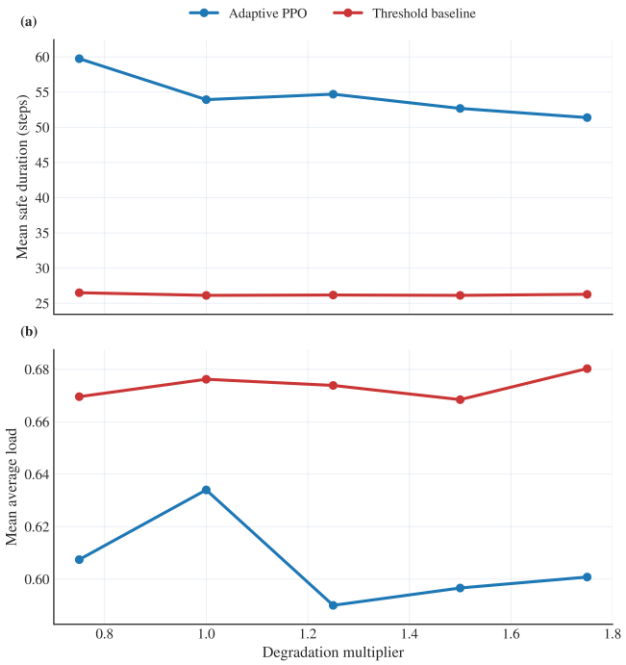


Figure 4. System performance under different subsystem degradation rates.

Policy Response Analysis

To clarify the decision-making logic of the learned controller, we examined how the policy responds to local perturbations in individual state variables while holding the remaining states fixed. The analysis shows that the supplemental water action increases sharply as the water state approaches its lower safety bound, whereas recycling intensity becomes more aggressive when subsystem reliability declines. In contrast, the policy suppresses energy-intensive regulation when the energy reserve is already limited.

Discussion

Compared with threshold-based control, the learned policy exhibits earlier adjustment and lower cumulative environmental burden in the tested stochastic scenarios.

Another important implication of the results is that subsystem reliability should not be treated as an external background factor. In the framework, reliability is explicitly coupled with recycling efficiency and ecological load evolution.

The same control architecture can be extended to additional modules such as waste processing, thermal regulation, or carbon dioxide control.

Offline training should be distinguished from onboard execution. In this study, PPO training used an RTX 4090 GPU, but onboard deployment would require policy inference only rather than policy optimization. Because the state and action spaces are compact, the inference burden is expected to be much lighter than the training burden. Practical deployment would still require explicit benchmarking of latency, memory footprint, and power consumption on target avionics platforms, and periodic offline retraining is more realistic than continuous onboard updating.

At the same time, several limitations should be acknowledged. The current state-transition model is intentionally simplified in a bid to make policy learning tractable within a compact framework. Some physical processes, ranging from detailed subsystem latency, maintenance scheduling to nonlinear cross-coupling between environmental variables, are not explicitly modeled. In addition, the present validation remains limited to synthetic disturbance scenarios and simulated subsystem degradation. As a result, the reported gains should be interpreted as evidence of control potential under a structured stochastic model, rather than as a direct guarantee of performance in operational space habitats. Stronger engineering claims would require validation against telemetry-derived disturbance profiles, subsystem-specific degradation data, and hardware-in-the-loop execution tests.

Conclusions

This study developed a PPO-based regulation framework for micro closed-loop space station ecosystems and evaluated it under stochastic simulation with reliability decay and environmental coupling. Within this setting, the learned policy maintained resource safety margins and improved regulation smoothness and long-term ecological performance relative to threshold-based control. These results provide simulation-level evidence of control feasibility, while telemetry-informed validation and hardware-in-the-loop testing remain necessary for engineering deployment.

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Declaration of AI-Assisted Technologies

During the preparation of this manuscript, AI-assisted tools were used only for grammar checking and language refinement. All modeling, algorithm design, simulation development, result interpretation, and scientific conclusions were independently completed and verified by the authors. The authors take full responsibility for the integrity and originality of the content.

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